

DANIEL LARSON

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SUMMARY

Statistics and Data Science student (UCSB) building ML pipelines, NLP systems, and forecasting models across sports analytics and startup settings; delivers measurable outcomes from raw data using Python, R, and SQL.

EDUCATION

University of California, Santa Barbara <i>B.S. in Statistics and Data Science GPA: 3.50</i> Relevant Courses: Principles of Data Science, Regression Analysis, Applied Stochastic Processes, Statistical Learning, Design and Analysis of Experiments, Time Series, Big Data Analysis, STAT Data Science.	Santa Barbara, CA Sep. 2024 – Dec. 2026
College of the Canyons <i>Completed IGETC certified coursework General Chemistry I & II, Statistics, Calculus I & II</i>	Santa Clarita, CA Aug. 2021 – Jun. 2024

EXPERIENCE

Tenart Tennis Platform Operations Intern <i>Shanghai Buba Sports & Culture Co., Ltd. (Tenart)</i> - Achieved 100% data integrity by building a SQLite database pipeline that integrated the Feishu API with platform trend logging, backed by a custom validation framework. - Informed Tenart's listing optimization strategy by engineering a Python web scraper (Selenium, undetected-chromedriver, BeautifulSoup) and SQLite-backed competitive intelligence system that surfaced 500+ reviews across 162 Shanghai tennis venues, quantifying gaps such as parking friction (cited 324 times across 40 venues). - Segmented competitor venues into validated review-theme clusters by applying Chinese NLP (jieba tokenization, TF-IDF vectorization) and K-Means, confirming cluster stability via bootstrap-resampled Adjusted Rand Index. - Replaced unvetted AI research with validated industry consensus by publishing bilingual playbooks that synthesized 27 primary sources on platform algorithm behavior across Douyin, Xiaohongshu, and Dianping. - Projected tennis search demand trends by building an ARIMA time-series forecasting model (ADF stationarity testing, ACF/PACF order selection, AIC-based comparison) on 35+ weeks of Baidu Index data, and grew search visibility via an 85-term keyword matrix across 5 customer personas and 3 platforms. - Translated trend analysis and competitive benchmarks into weekly bilingual (Chinese/English) KPI reports for the founder, while operating across both on-court coaching and data science functions in a fast-paced startup environment.	Jun. 2026 – Aug. 2026 Shanghai, China
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PROJECTS

SpiceRack – Python, Flask, Scikit-learn, NLP, TF-IDF, K-Means, SQLite, REST APIs - Shipped a recipe recommendation web app on a fixed showcase deadline by leading a cross-functional team as project manager across ML, backend, and frontend tracks. - Achieved a 0.78 silhouette score across ~475 clusters on 2.2M+ recipes by designing a custom NLP pipeline (TF-IDF with ID² boosting → TruncatedSVD → MiniBatchKMeans) that resolved cluster dominance and sparse-input failure modes. - Cut recommendation latency and enabled seamless AJAX interactions by engineering a top-K cosine similarity search with thread-safe REST API image caching and a SQLite-backed saved recipes system. - Placed 3rd out of 100+ teams at the UCSB Data Science Club Project Showcase, validating end-to-end ML deployment from raw data to production web app.	Oct. 2025 – May. 2026
MLB Pitcher Injury Risk Prediction – R, XGBoost, K-Means, tidymodels - Predicted next-season MLB pitcher injury risk by building and validating 4 ML models (logistic regression, elastic net, random forest, XGBoost) on 30 years of Lahman Database pitching data, identifying total innings pitched – including postseason – as the leading risk factor. - Revealed workload as a key injury driver by segmenting pitchers into 3 archetypes via K-means clustering; the highest-workload group carried nearly double the injury rate (26.6%) of the lowest (14.1%). - Tuned the final model to correctly flag 80% of at-risk pitchers, with the top 10% highest-risk group injured at a 53.6% rate – nearly 3x the 19% baseline average.	Jan. 2026 – Mar. 2026

ACTIVITIES

Data Science Club (UCSB) Won “Best Sports Analyst Project” at Datathon 2025; named a 2-time Project Pitchfire Finalist (SpiceRack, Anti-DUI) among competing student teams.	Oct. 2024 – Present
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SKILLS

Programming Languages: Python, R, SQL, C++ Concepts: Machine Learning, Statistical Modeling, Classification, Clustering, Regression, Natural Language Processing, Tokenization, Time Series Forecasting, Computer Vision, Feature Engineering, A/B Testing, REST APIs Libraries & Frameworks: Scikit-Learn, XGBoost, tidymodels, statsmodels, Flask, Pandas, NumPy, OpenCV, Selenium, BeautifulSoup, jieba Tools: Microsoft Excel, Git, GitHub, Jupyter Notebook, VS Code, Docker, SQLite, Feishu API Languages (Spoken): English (Native), Spanish (Elementary), Mandarin Chinese (Beginner) Achievements: Dean's Honors (L&S)
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